# Week 12

**Course Level Objectives:**

- Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. (CLO 5)
- Describe and compute confidence intervals. (CLO 6)
- Perform hypothesis testing using statistical methods. (CLO 8)

| Module Objectives | Instructional Materials | Learning Activities | Assessments | Tools |
|---|---|---|---|---|
| **1.** Explain why inference for a mean uses the t distribution with df = n − 1. (CLO 5)<br>**2.** Check the conditions for t procedures. (CLO 5)<br>**3.** Construct and interpret a t confidence interval for a population mean. (CLO 6)<br>**4.** Perform a one-sample t test and state the conclusion in context. (CLO 8) | e-Text: 6.2-CI Confidence Interval for a Mean<br><br>e-Text: 6.2-D Distribution of a Mean<br><br>e-Text: 6.2-HT Hypothesis Test for a Mean<br><br>Interactive Lecture: 6.2 Inference for a Mean<br><br>Slides: 6.2 Inference for a Mean<br><br>StatKey – Theoretical Distributions (t) | Participation: Inference for a Mean – gribbles CI and Chips Ahoy test worked in groups; Your Turn: Testing a Mean (juice bottles underfilled?) | Online Homework: 6.2 Inference for a Mean (due Sun Nov 15, 11:59 PM)<br><br>Take-Home Quiz 7 (due Mon Nov 16, 11:59 PM) – covers 6.1–6.3 (all objectives)<br><br>Test 3 (Fri Nov 20 – Sun Nov 22, proctored) – covers 4.4–6.5 (all objectives)<br><br>Test 3 Study Guide (all objectives)<br><br>Practice Test 3 (optional; due Thu Nov 19) (all objectives) | WileyPLUS<br>StatKey<br>Blackboard<br>Respondus |